ANU TRIPATHI

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EDUCATION

University of Minnesota-Twin Cities, Ph.D.in Civil Engineering 2015 - 2020

Indian Institute of Technology Kanpur, B.Tech.-M.Tech. in Civil Engineering 2010 - 2015

RESEARCH INTEREST

Finite element modeling, Injury biomechanics, Biological digital twins, Human-machine interaction, Wearable sensors, Signal processing, Damage mechanics.

GRANTS PROPOSALS

1. Office of Naval Research (ONR) (Submitted) 2026
   - **Tripathi, A. (PI)** and Carlsen, R.W. (co-PI)
   - 'Human digital twins for injury risk prediction under prolonged repetitive loading in military-relevant conditions'

2. Office of Naval Research (ONR) (Funded) 2024 - 2025
   - Carlsen, R.W. (PI) and **Tripathi, A. (Co-I)**
   - 'Development of human digital twins for injury risk prediction under military-relevant loading'

RESEARCH EXPERIENCE

Postdoctoral Researcher
Robert Morris University, Greater Pittsburgh Area 2023 - present
University of Wisconsin-Madison 2021 - 2023
Advisors: Prof. Rika Carlsen, Prof. Christian Franck

1. **Youth soccer mild traumatic brain injury (mTBI) risk assessment** 2021 - present
   – Created anatomically detailed subject-specific human head digital twins from medical images.
   – Incorporated in-vivo brain material properties from magnetic resonance elastography (MRE) in the human head FE model for subject-specific material properties.
   – Automated the pipeline for FE head model validation against experimental datasets to advance model standardization efforts.
   – Developed a user-friendly wearable instrumented headband and data-processing technique to record accurate head kinematics in soccer.
   Collaborators: Dr. Peter Ferrazzano, Dr. Alison Brooks, Prof. Haneesh Kesari, Prof. Joseph Andrews, Prof. Phil Bayly, Dr. Chad Hovey

2. **Development of a whole body human digital twin for injury risk assessment in military-relevant conditions** 2024 - present
   – Developed novel FE-based mesh morphing to incorporate the human head FE model into an FE body model for an integrated human digital twin for injury risk prediction.
   Collaborators: Prof. Daniel Nicolella (Southwest Research Institute)

3. **High resolution optic nerve modeling** 2021 - 2022
   – Built custom tensile testing setup to quantify mechanical properties of fresh soft biological tissue.
   – Developed micro-CT based FE models of porcine optic nerves for multi-scale assessment of traumatic optic nerve injury.
   Collaborators: Prof. Walter Schneider (University of Pittsburgh)

Graduate Research Assistant 2015 - 2020
University of Minnesota Twin Cities
Chemo-mechanical modeling of stress corrosion cracking in semi-crystalline polymers
Advisors: Prof. Jia-Liang Le, Prof. Susan Mantell
– Developed damage constitutive model of high density polyethylene, a semi-crystalline polymer.
– Coupled the diffusion-chemical and damage mechanical models of polymer for life-time estimation of pipes in chlorinated water.

Graduate Research Assistant 2014 - 2015
IIT Kanpur, India
Seismic Vulnerability Assessment of Stone Masonry Monasteries in Sikkim Himalayas
Advisor: Prof. Durgesh C. Rai
– Built FE model of stone masonry structures to quantify their damage response under seismic loading.

Research Assistant May - Jul 2013
TU Kaiserslautern, Germany
Design of Fiber Reinforced Polymers as Reinforcement and Connectors in Sandwich Panel
Advisor: Jun. Prof. Dipl. Ing. Matthias Pahn
– Performed deflection analyses of glass fiber reinforced polymers reinforced slabs under flexure.

JOURNAL PUBLICATIONS

Published/Accepted

1. **Tripathi, A.**, Brooks, A., Snedden, T.R., Ferrazzano, P., Franck, C., Carlsen, R.W., 2026. Effect of individual differences in cortical folds on brain deformation from head accelerations: A computational study. *Biomechanics and Modeling in Mechanobiology*.
2. **Tripathi, A.**, Wan, Y., Zhu, Z., Camci, F., Turcsanyi, S., Kachhadiya, J. P., Canizales, M.A., Brooks, A., Kesari, H., Andrews, J., Snedden, T.R., Ferrazzano, P., Franck, C., Carlsen, R.W., 2026. Field evaluation of a wearable instrumented headband designed for measuring head kinematics. *Annals of Biomedical Engineering*.
3. **Tripathi, A.**, Wan, Y., Malave, S., Turcsanyi, S., Fawzi, A.L., Brooks, A., Kesari, H., Snedden, T.R., Ferrazzano, P., Franck, C., Carlsen, R.W., 2025. Laboratory evaluation of a wearable instrumented headband for rotational head kinematics measurement. *Annals of Biomedical Engineering*, pp.1-18.
4. **Tripathi, A.**, Mantell, S. and Le, J.L., 2021. Chemo-mechanical modeling of static fatigue of high density polyethylene in bleach solution. *International Journal of Solids and Structures*, 217, 90-105.
5. **Tripathi, A.**, Mantell, S. and Le, J.L., 2019. A Morphology Based Constitutive Model for High Density Polyethylene. *Mechanics of Materials*, p.103091.
6. **Tripathi, A.**, and Rai, D.C., 2019. Seismic vulnerability assessment and fragility analysis of stone masonry monastic temples in Sikkim Himalayas. *International Journal of Architectural Heritage*, 13(2), pp.257-272.
7. **Tripathi, A.**, Mantell, S. and Le, J.L., 2017. Modeling of cohesive fracture interacting with a stationary capillary fluid. *Engineering Fracture Mechanics*, 182, pp.19-31.
8. Rai, D.C., Singhal, V., Pradhan, T. and **Tripathi, A.**, 2016. Seismic vulnerability of monastery temples of stone masonry in Sikkim Himalaya. *Current Science*, 110(10), pp.1947-1957.

Under Review/Submitted

1. Lewis, C., **Tripathi, A.**, Carlsen R.W., Head Kinematics and Brain Tissue Deformation from Soccer Heading: A Review of Implications for Brain Injury Risk. *Annals of Biomedical Engineering*. arXiv. (Submitted June, 2026)

2. Cooper, C., **Tripathi, A.**, Palardy, G., Upadhyay K., History Matters: Damage-Mediated Amplification of Brain Deformation and Injury Risk under Repeated Head Impacts. *Biomechanics and Modeling in Mechanobiology*. (Submitted Aug, 2026)

Under Preparation

1. Tzoumaka, A., **Tripathi, A.**, Wang, Y., Fawzi, A. L., Franck, C., Carlsen R.W., Rate Inclusive Strain Estimate (RISE) for Brain Injury Prediction. *Annals of Biomedical Engineering*. (Under preparation, to be submitted in Oct 2026)
2. Tzoumaka, A., **Tripathi, A.**, Carlsen R.W., Finite element based mesh morphing for bio-fidelic brain model development. *Computer Methods in Biomechanics and Biomedical Engineering*.
3. **Tripathi, A.**, Upadhyay, K., Franck, C., Carlsen R.W., AI-driven identification of head acceleration event type in youth soccer.

CONFERENCE PRESENTATIONS

1. **Tripathi, A.**, Buche, M., Hovey, C., Lejune, E., Carlsen, R.W., 2024. Automated Workflow for a Subject-Specific Human Digital Twin. *International Mechanical Engineering Congress and Exposition (IMECE), ASME*, Portland, OR, USA.
2. **Tripathi, A.**, Gonzalez, J. G., Snedden, T., Brooks, A., Ferrazzano, P., Franck, C., Carlsen, R. W., 2024. Effect of Modeling Cortical Folds on Brain Deformation From Head Accelerations: A Computational Study. *The Summer Biomechanics, Bioengineering and Biotransport Conference (SB3C), ASME*, Lake Geneva, WI, USA.
3. **Tripathi, A.**, Wang, Y., Lejune, E., Carlsen, R.W., Terpsma, R., Hovey, C., 2023. Towards a fully automated workflow for subject-specific human digital twin for traumatic brain injury risk assessment. *International Mechanical Engineering Congress and Exposition (IMECE), ASME*, New Orleans, LA, USA.
4. **Tripathi, A.**, Gonzalez, J. G., Snedden, T., Brooks, A., Ferrazzano, P., Franck, C., Carlsen, R. W., 2022. Effect of Individual Anatomical Differences in Brain Deformation From Head Accelerations: A Computational Study. *International Mechanical Engineering Congress and Exposition (IMECE), ASME*, Columbus, OH, USA.
5. **Tripathi, A.**, Mantell, S., Le, J-L., 2020. Reduced order kinetics model for corrosion of polyethylene in bleach solution. *77th Annual Technical Conference and Exhibition of the Society of Plastics Engineers, SPE ANTEC, Society of Plastics Engineers*, San Antonio, TX, USA. (The Virtual Edition).
6. **Tripathi, A.**, Mantell, S., Le, J-L., 2018. Morphology dependent damage constitutive model for ductile and brittle failure of semi-crystalline polymers. *International Mechanical Engineering Congress and Exposition (IMECE), ASME*, Pittsburgh, PA, USA.
7. **Tripathi, A.**, Rai, D.C., 2018. Seismic fragility analyses of stone masonry monastic temples in Sikkim. In *11th National Conference on Earthquake Engineering*, Los Angeles, CA, USA.
8. **Tripathi, A.**, Mantell, S., Le, J-L., 2017. Modeling of Cohesive Crack Interaction with Stationary Capillary Fluid. *Engineering Mechanics Institute Conference*, San Diego, CA, USA.

POSTERS

1. Tzoumaka, A., **Tripathi, A.**, Carlsen, R.W., 2025. Optimizing Head Modeling for TBI Research: A Comparison of Mesh Morphing Techniques. *American Society of Biomechanics* Pittsburgh, PA, USA.
2. **Tripathi, A.**, Wang, Y., Carlsen, R.W., Lejune, L., Terpsma, R., Hovey, C., 2025. Towards a fully automated workflow for subject-specific human digital twin for traumatic brain injury risk assessment. *Military Health System Research Symposium (MHSRS)* Kissimmee, FL, USA.
3. Roebuck, I., **Tripathi, A.**, Tzoumaka, A., Carlsen, R.W., 2025. Optimizing Head Modeling for TBI Research: A Comparison of Mesh Morphing Techniques. *BioE Day, University of Pittsburgh* Pittsburgh, PA, USA.

4. Pfuma, T., **Tripathi, A.**, Snedden, T., Brooks, A., Ferrazzano, P., Franck, C., Carlsen, R.W., 2025. Effect of Header Type and Sex on Rotational Head Kinematics of Youth Players During Soccer Heading. *BioE Day, University of Pittsburgh* Pittsburgh, PA, USA.
5. **Tripathi, A.**, Carlsen, R.W., 2024. Development of a whole-body human digital twin for injury risk assessment in military-relevant conditions. *PANTHER Workshop* Madison, WI, USA.
6. **Tripathi, A.**, Malave, S., Wan, Y., Canizales, M.A., Zhu, Z., Fawzi, A.L., Turcsanyi, S., Brooks, A., Kesari, H., Ferrazzano, P., Snedden, T.R., Franck, C., Carlsen, R.W., 2023. An instrumented headband wearable device for head kinematics measurement in soccer. *Biomedical Engineering Society Annual Meeting* Seattle, WA, USA.
7. **Tripathi, A.**, Wang, Y., Carlsen, R.W., Lejune, L., Terpsma, R., Hovey, C., 2023. Towards a fully automated workflow for subject-specific human digital twin for traumatic brain injury risk assessment. *Military Health System Research Symposium (MHSRS)* Kissimmee, FL, USA.
8. **Tripathi, A.**, Nakarmi, S., Wang, Y., Carlsen, R.W., Sokolow, A., Terpsma, R., Hovey, C., 2022. Demonstration of a semi-automated workflow for subject-specific human digital twin. *Military Health System Research Symposium (MHSRS)* Kissimmee, FL, USA.
9. **Tripathi, A.**, Wu, Y., Schneider, W., Carlsen, R.W., 2022. High-resolution MR imaging and finite element modeling of porcine optic nerve under tensile stretch. *Military Health System Research Symposium (MHSRS)* Kissimmee, FL, USA.
10. **Tripathi, A.**, Nakarmi, S., Wang, Y., Carlsen, R.W., 2022. Demonstration of a semi-automated workflow for subject-specific human digital twin. *BioE Day, University of Pittsburgh*, PA, USA.
11. **Tripathi, A.**, 2019. Morphology-dependent damage constitutive model for ductile and brittle failure of semi-crystalline polymers. *3M Panel Session, University of Minnesota*, MN, USA.

CONFERENCE PROCEEDINGS

1. **Tripathi, A.**, Le, J.L., and Mantell, S., 2020. Reduced order kinetics model for corrosion of polyethylene in bleach solution. In *77th Annual Technical Conference and Exhibition of the Society of Plastics Engineers, SPE ANTEC, Society of Plastics Engineers*, San Antonio, TX, USA.
2. **Tripathi, A.** and Rai, D.C., 2018. Seismic fragility analyses of stone masonry monastic temples in Sikkim. In 11th National Conference on Earthquake Engineering, Los Angeles, CA, USA.
3. Rai, D.C., Tomar, A.S., **Tripathi, A.**, 2015. Recreated Masonry Sub Assemblages for Characterization of the Existing Masonry. In *12th North American Masonry Conference*, Denver, CO, USA.

TEACHING EXPERIENCE

Instructor Spring 2024
Robert Morris University
ENGR 2510: Introduction to Biomedical Engineering

Teaching Assistant Fall 2018, Spring 2019
University of Minnesota-Twin Cities
CE 4401: Steel and Concrete Design, Prof. Catherine French

Tutor Summer 2019
University of Minnesota-Twin Cities
College of Science and Engineering
Teaching Training for New International Teaching Assistants

Teaching Assistant Spring 2015
Indian Institute of Technology Kanpur, India
CE 242: Construction Materials, Prof. Sudhir Mishra

Teaching Assistant Fall 2014
Indian Institute of Technology Kanpur, India
CE 251: Structural Analysis, Prof. Samit Ray Chaudhary

TECHNICAL SKILLS

Patient-specific Finite Element (FE) Modeling:
Medical image processing: FreeSurfer, 3D Slicer, ANTs, Mimics, MATLAB;
FE meshing: Hypermesh, CUBIT;
FE simulations: ABAQUS (10+ years), LS-DYNA (1 year);
FE post-processing: Abaqus-Python;
Other:
Signal Processing: Mathematica (Continuous Wavelet Transform-based adaptive filtering);
Material modeling: Abaqus (VUMAT, UMAT, and HETVAL);
Statistical/ML analyses: MATLAB, Python.

SERVICE AND PROFESSIONAL ACTIVITIES

1. Undergraduate Student Mentor 2021 - Present
   - Mentored 15+ students in their undergraduate research
   - Mentored Careers Undergraduate research scholars and RMU presidential scholars on their honors thesis
2. Reviewer
   - Frontiers in Bioengineering and Biotechnology 2025 - present
   - Journal of Applied Biomechanics 2025 - present
   - Summer Biomechanics, Bioengineering, and Biotransport Conference 2025 - present
   - American Society of Biomechanics Conference 2025 - present
   - Biomedical Engineering Society Conference 2022 - 2023
   - Journal of Civil Engineering, Korean Society of Civil Engineers 2017 - 2019
3. Earthquake Engineering Research Institute (EERI)
   - President, EERI at the University of Minnesota 2018 - 2020
4. Society of Women Engineers (SWE)
   - Graduate Student Mentor 2017 - 2018
5. Current Professional Memberships
   - Biomedical Engineering Society (BMES)
   - American Society of Mechanical Engineers (ASME)
   - American Society of Biomechanics (ASB)

OTHER GRANTS AND FELLOWSHIPS

1. Sommerfeld Fellowship 2015 - 2016
   University of Minnesota-Twin Cities, MN
2. Working Internship in Science and Engineering (WISE) May - Aug 2013
   Deutscher Akademischer Austauschdienst (DAAD), Germany
3. ACCESS/XCEED Bridges Support, Pittsburgh Supercomputer Center (PSC) 2021-Present
   Computational Service Hours on the supercomputer facilities (Equivalent to $\$10k$)